



L-31 (3/3; HT)

Oct 06, 2026

Single-Sample Z-Test: Example 1

- A random sample of $n = 100$ recorded deaths in the United States during the past year showed an average life span of $\bar{x} = 71.8$ years
- Assume a population standard deviation of $\sigma = 8.9$ years
- Does this seem to indicate that the mean life span today is greater than 70 years? Use a $\alpha = 0.05$ level of significance

$$H_0: \mu = 70$$

versus

$$H_1: \mu > 70$$

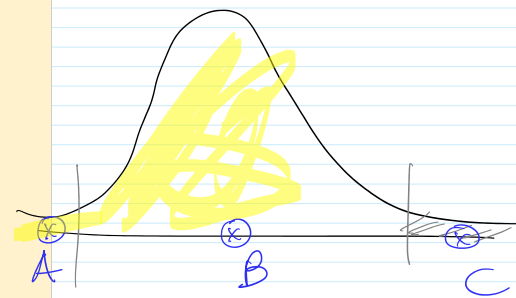
• Test statistic: Z-score

• Test-Type: Right-Tail (critical value) $Z_{\alpha} = Z_{0.05} = 1.645$

• Test score: $Z_{calc} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{71.8 - 70}{8.9 / \sqrt{100}} \approx 2.02$

• Decision: Since $|Z_{calc}| > |Z_{\alpha}| \Rightarrow \text{Reject } H_0$

• Consequence: Mean Life span is > 70 years @ 0.05 significance level



Single-Sample Z-Test: Example 2

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- A new synthetic fishing line is claimed to have a mean breaking strength of $\mu = 8 \text{ kg}$ with a standard deviation of $\sigma = 0.5 \text{ kg}$
- A random sample of $n = 50$ lines is tested and found to have a mean breaking strength of $\bar{x} = 7.8 \text{ kg}$
- Test $H_0: \mu = 8 \text{ kg}$ against $H_1: \mu \neq 8 \text{ kg}$ at the $\alpha = 0.01$ level of significance

Two-Sample Z-Test: Example 1

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- Two types of steel cables are tested for tensile strength
- Brand A: $n_1 = 40$ cables, sample mean strength $\bar{x}_1 = 108 \text{ kN}$
- Brand B: $n_2 = 50$ cables, sample mean strength $\bar{x}_2 = 104 \text{ kN}$
- Past manufacturing records give population stdev $\sigma_1 = 8 \text{ kN}$; $\sigma_2 = 10 \text{ kN}$
- At $\alpha = 0.05$, test if Brand A has a significantly higher mean tensile strength than Brand B

- $H_0: \mu_A - \mu_B = 0$ vs. $H_1: \mu_A - \mu_B > 0$
 standard error in $(\bar{x}_1 - \bar{x}_2)$, $SE = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2} \approx 1.90 \text{ kN}$
 Test statistic, Z-score
 Test type right-tail. Critical value $Z_{0.05} \approx 1.645$
 Test score $Z_{\text{calc}} = \frac{108-104}{1.90} \approx 2.11$
 Decision: Since $|Z_{\text{calc}}| > |Z_{\text{crit}}| \Rightarrow \text{Reject } H_0$
 Consequence: Brand A has significantly higher mean tensile strength than Brand B

$$Y = aX_1 + bX_2$$

$$\sigma_Y^2 = a^2 \sigma_1^2 + b^2 \sigma_2^2$$

Two-Sample Z-Test: Example 2

- A chemical process was run using two different catalysts
- Catalyst 1: $n_1 = 36$ runs, average yield 85%
- Catalyst 2: $n_2 = 40$ runs, average yield 81%
- Population standard deviations are known: $\sigma_1 = 4\%$ and $\sigma_2 = 5\%$

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- Catalyst 1: $n_1 = 36$ runs, average yield 85%
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- Population standard deviations are known: $\sigma_1 = 4\%$ and $\sigma_2 = 5\%$
- Test $H_0: \mu_1 - \mu_2 = 0$ against $H_1: \mu_1 - \mu_2 \neq 0$ at a $\alpha = 0.01$ level of significance

$$\text{std. error, } SE = \sqrt{(4\%)^2/36 + (5\%)^2/40} \approx 1.03\%$$

$$Z_{\text{crit}} = Z_{0.005} \approx 2.576$$

$$Z_{\text{calc}} \approx \frac{85\% - 81\%}{1.03\%} \approx 3.87$$

$$|Z_{\text{calc}}| > |Z_{\text{crit}}| \quad \text{Reject } H_0$$

Single-Sample t-Test: Example 1

- A civil engineer tests $n = 12$ concrete cylinders produced under a new curing process. The sample mean 28-day compressive strength $\bar{x} = 38.5$ MPa, sample standard deviation $s = 2.1$ MPa
- The standard baseline requirement is 40 MPa
- At $\alpha = 0.05$, test whether the average strength is significantly lower than the required standard

Left-tailed T-test

$$\text{Critical Value, } t_{0.05, 11} = 1.796 \approx T_{\text{crit}}$$

$$\text{Test score} = \frac{T_{\text{calc}}}{(s/\sqrt{n})} = \frac{38.5 - 40}{(2.1/\sqrt{12})} \approx -2.47$$

$$|T_{\text{calc}}| > |T_{\text{crit}}| \Rightarrow \text{Reject } H_0 \quad \text{NOT } (n-1)$$

Single-Sample t -Test: Example 2

- A precision machine shop produces steel shaft pins designed to have a mean diameter of 10.00 mm
- A sample of $n = 10$ pins yields a mean diameter of $\bar{x} = 10.04$ mm and a sample standard deviation $s = 0.03$ mm
- Assume pin diameters are normally distributed
- Test $H_0: \mu = 10.00$ against $H_1: \mu \neq 10.00$ at $\alpha = 0.01$

Two-Sample t -Test: Example 1

- Two distinct asphalt mixtures are tested for rut depth (in mm) under identical load cycles
- Mix A: $n_1 = 10$, $\bar{x}_1 = 7.2$ mm, $s_1 = 1.1$ mm
- Mix B: $n_2 = 12$, $\bar{x}_2 = 8.5$ mm, $s_2 = 1.3$ mm
- Assume independent normal populations with equal variances
- Test whether Mix B experiences a significantly greater mean rut depth than Mix A at $\alpha = 0.05$

Two-Sample t -Test: Example 2

- Paired samples: to evaluate a structural retrofitting technique, 8 reinforced concrete columns are loaded to measure deflection (in mm) under stress, both before and after retrofitting
- Test if the retrofit significantly reduces mean deflection at $\alpha = 0.05$

Column	1	2	3	4	5	6	7	8
Before (X)	12.4	14.1	11.8	13.5	15.0	12.9	13.8	14.5
After (Y)	10.1	12.0	10.0	11.2	12.8	10.9	11.5	12.1

Variance χ^2 -Test: Example 1

- A quality control specification requires that the variance of rebar diameters must not exceed $\sigma_0^2 = 0.10 \text{ mm}^2$
- A random sample of $n = 21$ rebars yields a sample variance $s^2 = 0.18 \text{ mm}^2$
- Perform hypothesis test at $\alpha = 0.05$.

Variance χ^2 -Test: Example 2

- An automated scale must weigh cement bags with a variance $\sigma^2 = 0.04 \text{ kg}^2$
- A sample of $n = 15$ bags shows a sample variance of $s^2 = 0.08 \text{ kg}^2$
- Test whether the variance exceeds the specification limit at $\alpha = 0.025$

Variance χ^2 Goodness-of-Fit: Example 1

- An inspector records the location of cracks across 120 concrete slabs categorized into four quadrants (North, South, East, West)
- Observed counts: North = 38, South = 22, East = 34, West = 26
- Test the null hypothesis that cracks occur uniformly across all four quadrants at $\alpha = 0.05$

Variance χ^2 Goodness-of-Fit: Example 2

- A soil survey classifies 200 site samples into three soil types: Clay, Silt, and Sand
- The expected geological proportion is 2 : 2 : 1
- Observed counts: Clay = 90, Silt = 70, Sand = 40
- Test whether the observed distribution matches the expected distribution at $\alpha = 0.05$

Two-Variance F -Test: Example 1

- An engineer compares two automatic surveying tools to evaluate precision
- Tool A: $n_1 = 16$ measurements, $s_1^2 = 2.40 \text{ mm}^2$
- Tool B: $n_2 = 21$ measurements, $s_2^2 = 0.80 \text{ mm}^2$
- Test $H_0: \sigma_1^2 = \sigma_2^2$ against $H_1: \sigma_1^2 \neq \sigma_2^2$ at $\alpha = 0.05$

Two-Variance F -Test: Example 2

- A manufacturing plant uses two different suppliers for steel rods
- Supplier 1: $n_1 = 11$, $s_1 = 5.2$ MPa
- Supplier 2: $n_2 = 13$, $s_2 = 2.1$ MPa
- Test whether the variability in tensile strength for Supplier 1 is significantly greater than Supplier 2 ($H_0: \sigma_1^2 \leq \sigma_2^2$ vs $H_1: \sigma_1^2 > \sigma_2^2$) at $\alpha = 0.01$

Tests concerning mean

Test on variance

χ^2 goodness-of-fit

Kolmogorov-Smirnov (K-S) Test